Particulars of the Experiments Performed

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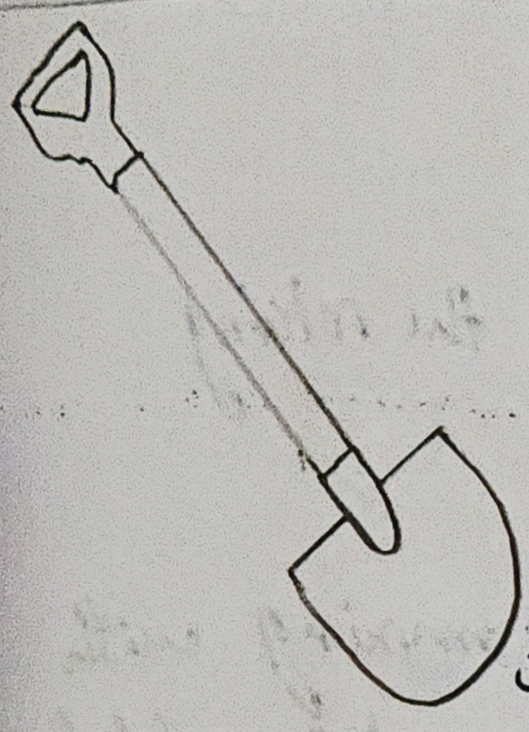
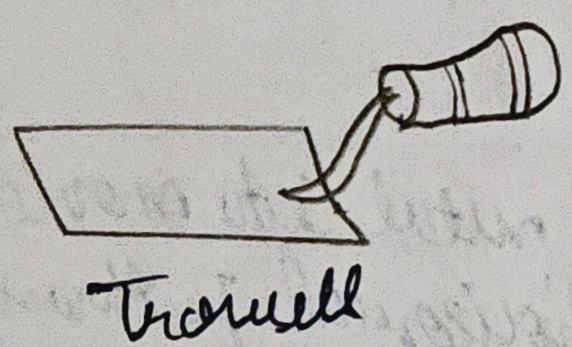
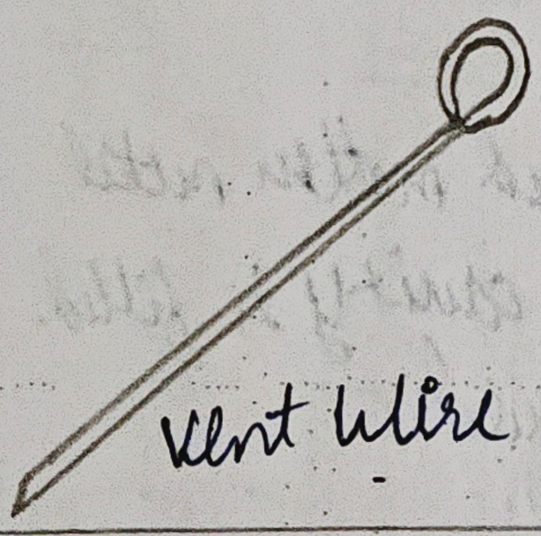
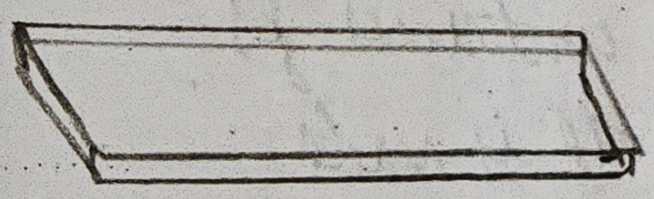
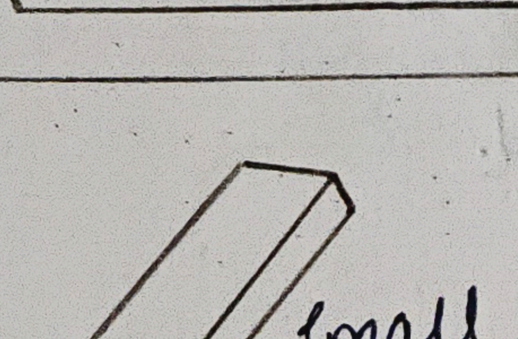
		Total			

CIE Marks (Reduced to 50)

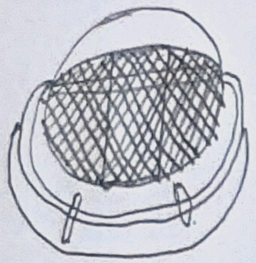
= {Total Class marks (Max.200) + Test marks (Max.50)}/ 50

Experiment 01

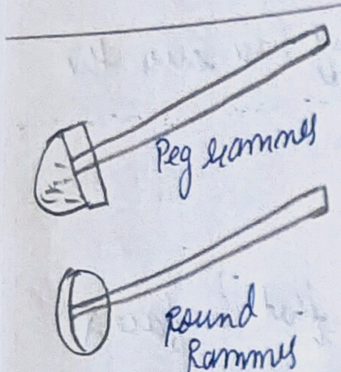
Sketches of tools - Foundry and their uses:

 Shovel	Shovel is used to loosen, pick and fill the sand into the moulding box over the moulding box.
 Trowell	These are used for finishing flat surface & corners of the mould. They are made with iron.
 Vent wire	This is thin steel rod with pointed edge and vent loop on other side after ramming and levelling.
Big level 	It is a flat bar made of wood to stick off the every sand from top of the box.
 Small level	similar to big level it's used to remove sand, and after big level is used. Provides us with smooth & flat surface
lifter	A lifter is a finishing tool used for repairing the mould & finishing the mould sand.

The main function of sieve is to remove the foreign material from the sand

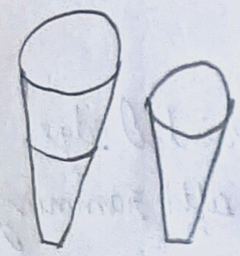


Sieve



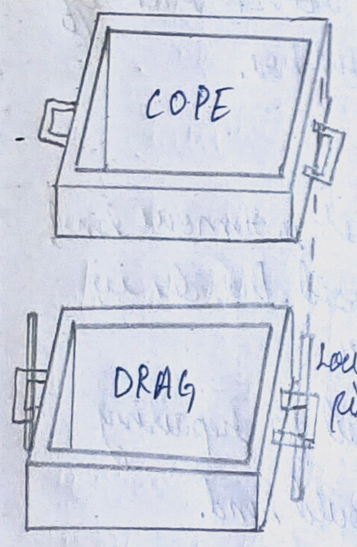
It is used to strike the sand mass & the m/cing box to pack the sand.

It is used for ramming once the ramming of peg rammer is completed & additional sand is added



Runner is used to pour the molten metal into mould cavity. The runner provides the horizontal pathway into mould cavity.

Rim is used to overflow the poured molten metal from the rim indicating that the cavity is filled. mould

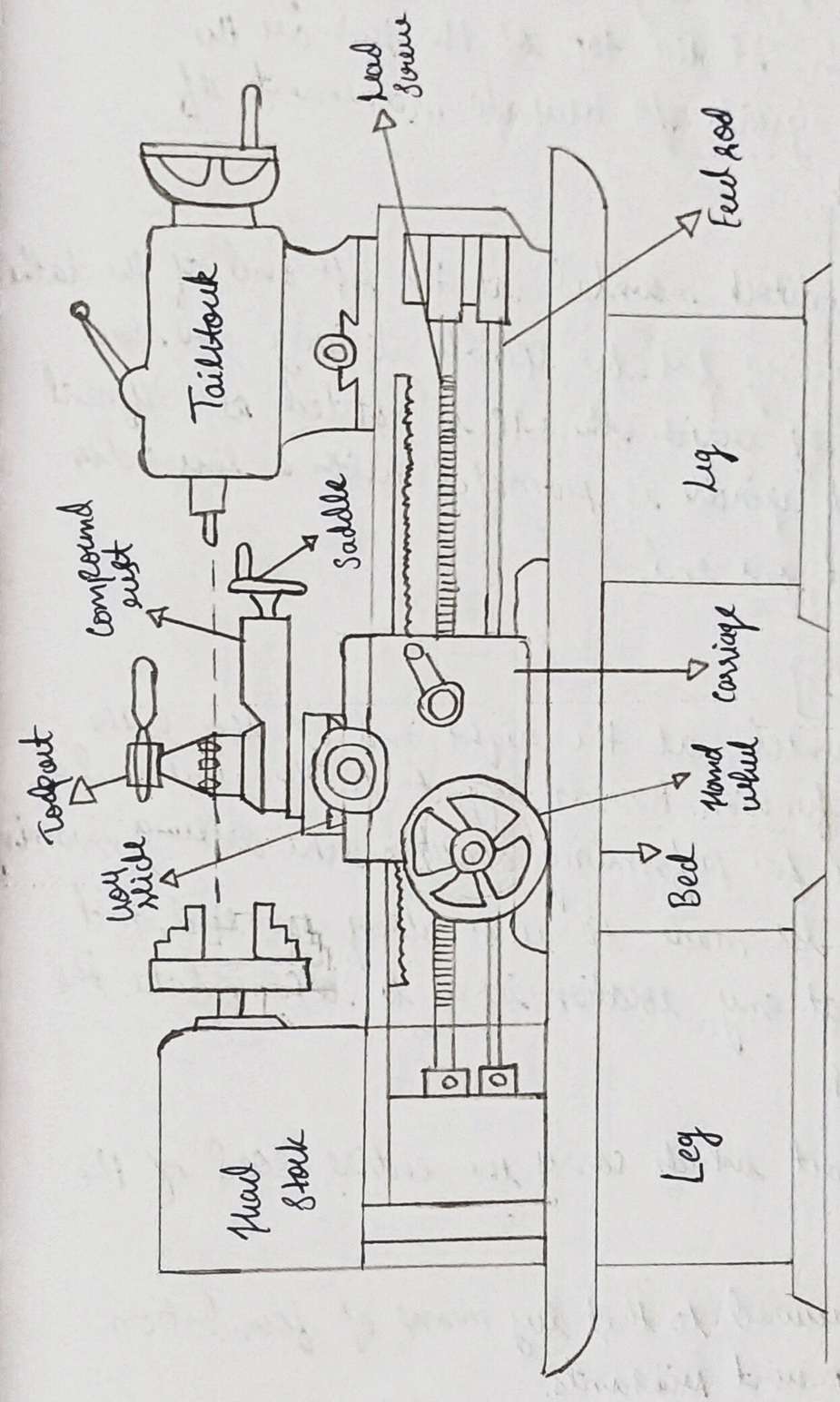


Moulding box is used as container for sand moulds to be made. It is made of iron & is open for 2 sides.

Staff Signature

Sketches of tools used in Machine Shop **Sketch:**

Engine Lathe



Description: Main parts of Engine lathe

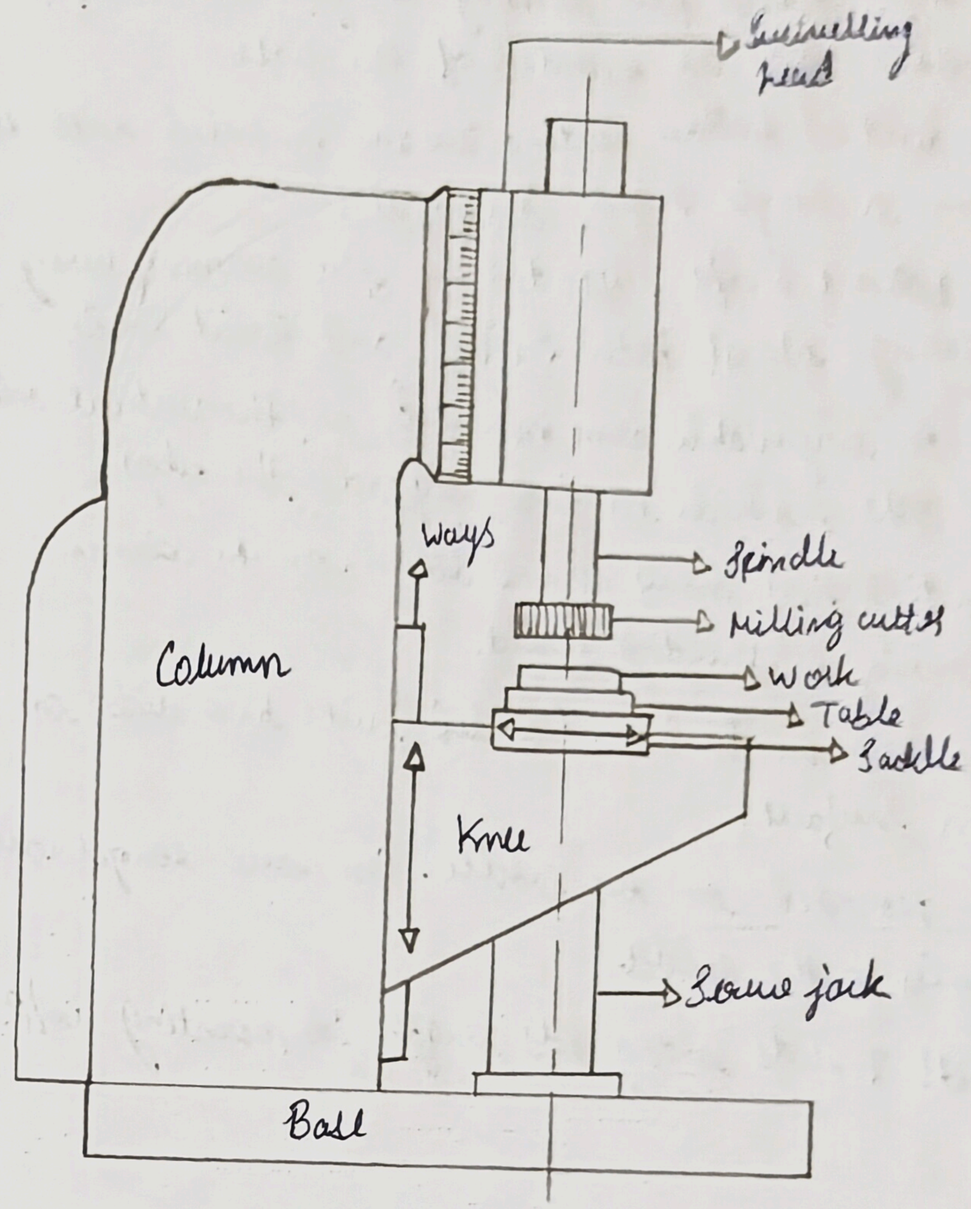
1) Bed :- It is a rigid structure which forms the base of foundation to support all the other parts such as headstock, tail stock, carriage etc. It is usually made from gray cast iron. At the top of the bed are the guideways, which guide for accurate movement of carriage and tailstock.

2) Headstock :- The headstock mounted at the left end of the bed serves as housing for the spindle deriving gear and pulleys by means of which the w/p is rotated at different speeds. The head stock spindle is provided with a live center or chuck to support one end.

3) Tailstock : [Dead center]
The tail stock mounted at the right end of the lathe bed performs two function. Provide support to the other end of w/p. Hold a tool for performing operation like drilling, reaming. The tailstock can be made to slide along the bed, and can be clamped at any location so as to accomodate the w/p at different lengths.

4) Legs : Are the support which carry the entire load of the machine above them.
They are firmly secured to the by means of foundation bolts in order to prevent vibrants.

Sketch: Column & Knee Type Milling Machine

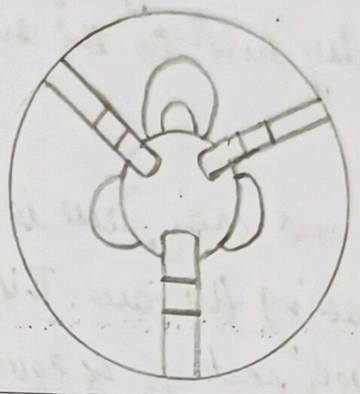


Description: Main parts of Column & Knee Type Milling Machine.

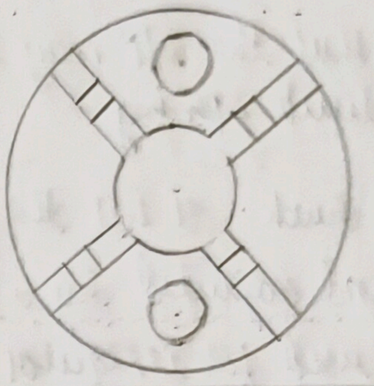
* Base :- A strong, hollow part forming the foundation of the m
All other machine part are mounted on the base.
* Column :- A vertical hollow casting. Usually combined with the
base to form a single rigid structure.
* Spindle :- A hollow shaft supported by the column using
bearing. Bearings absorb both radial and thrust loads.
* Arm :- An adjustable arm mounted on the vertical colum
Supports the yoke, which in turn supports the arbor.
* Knee :- A casting that moves up and down on the column's vert
guideways using an elevating screw.
* Saddle :- Mounted on the knee. Provided with two slides on its
and bottom surface.
* Worktable :- mounted on the saddle. Can move longitudinally
at right angles to the saddle.
* Provided with T-slots along its length for mounting w/p.

Sketches of lathe accessories:

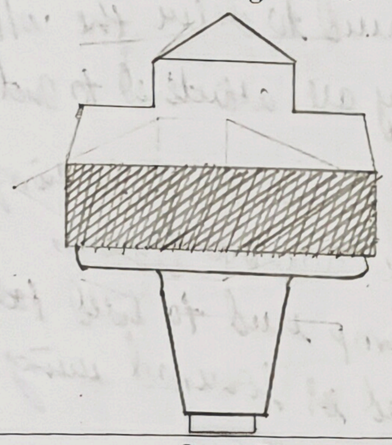
Three Jaw Chuck



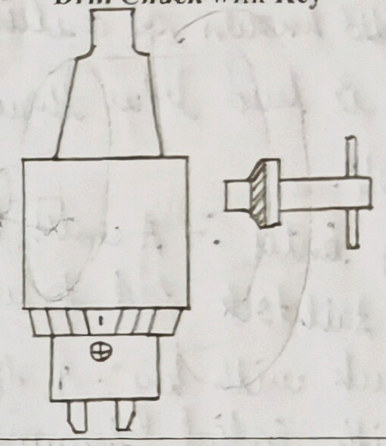
Four Jaw Chuck



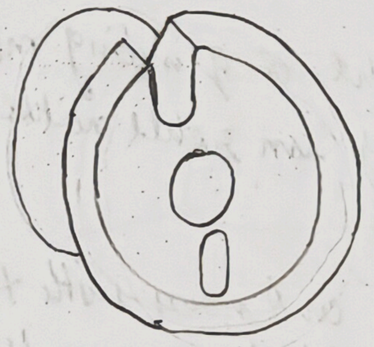
Revolving Centre



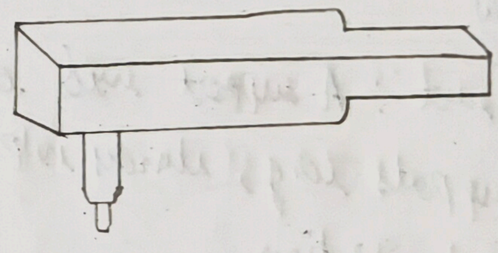
Drill Chuck with Key



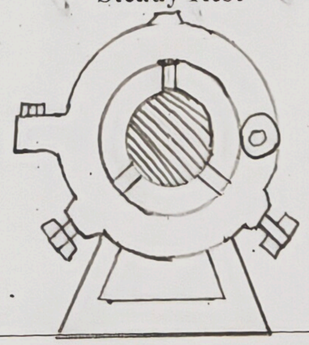
Catch Plate



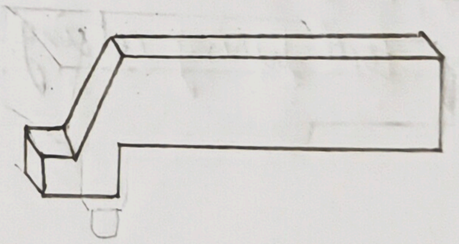
Carrier



Steady Rest



Follower Rest

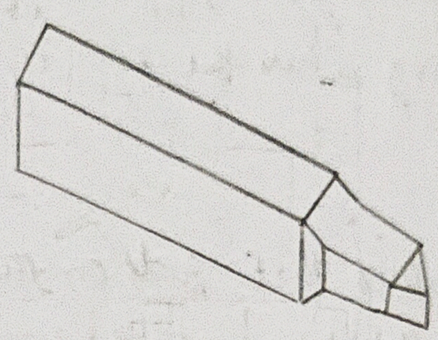


Description of Lathe Accessories:

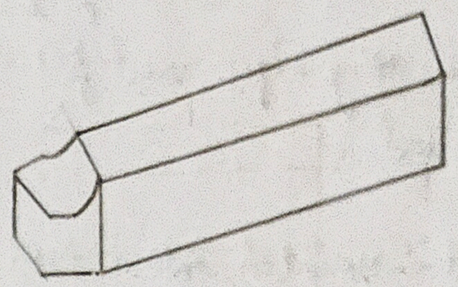
* Three jaw chuck :- All the three jaws simultaneously by turning a key & thus the w/p may automatically held in the centre of the chuck opening
* Four Jaw chuck :- It has the four jaws and each jaw is indepe actuated and adjusted by a key for reading the jaw. This type chuck is used for irregular shape, rough castings of square
* Carrier :- Also known as driving dogs and used to drive the w/p when it is held by the three centers. They are attached to end of the w/p.
* Revolving centre :- A centre that rotates in its own bearings at the tailstock end to support the work without friction
* Drill chuck with key :- A specialised clamp used to hold tool objects with radial symmetry, tightened or loosened using chuck key.
* Steady rest :- A support used on a lathe or grinding m/c, helps supports long, slender w/p so they can rotate without bending & vibration.
* Follows rest :- Mounted on the carriage and moves with the tool, used to prevent slender w/p from deflecting away from the cutting tool during m/cing.

Sketches of Cutting Tools:

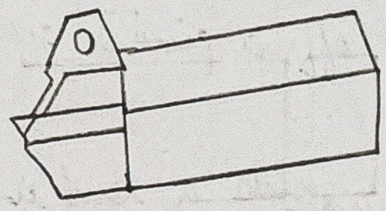
V Tool



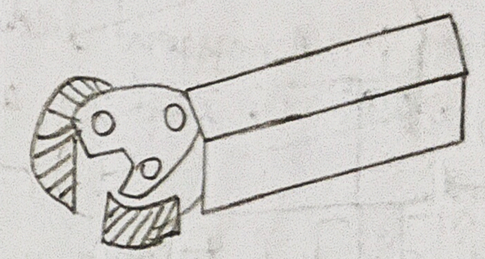
Facing Tool



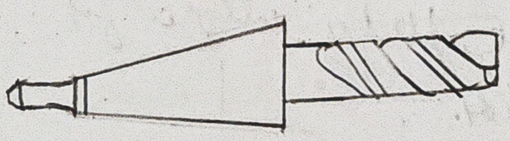
Parting off Tool



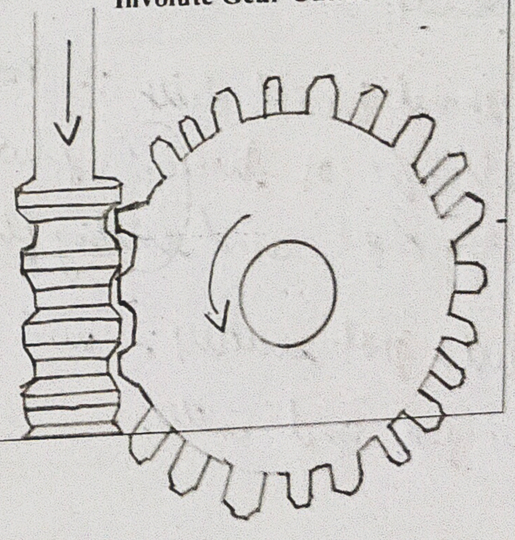
Knurling Tool



Taper Shank Twist drill



Involute Gear Cutter



Description of Cutting Tools:

* V Tool – A V-tool, chisel or gauge has a rectangular body with V-shaped cutting edges, used for carving where the edge is angled / curved.

* Facing tool :- Mounted into a tool holder that sits on the carriage of a lathe, used to feed the tool across the rotating w/p to produce a flat surface.

* Parting tool :- A narrow bladed tool used in turning/planning operation, used for cutting off a piece as a component.

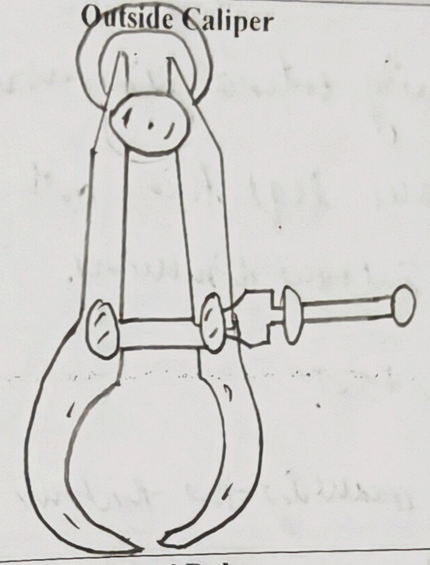
* Knurling tools :- used to press a pattern onto a round metal creates a textural grip, commonly seen on handles and screwdrivers.

* Taper shank twist drill :- Rotary cutting tool with one or more straight or helical flutes, flutes allow for passing of chips and cutting fluid.

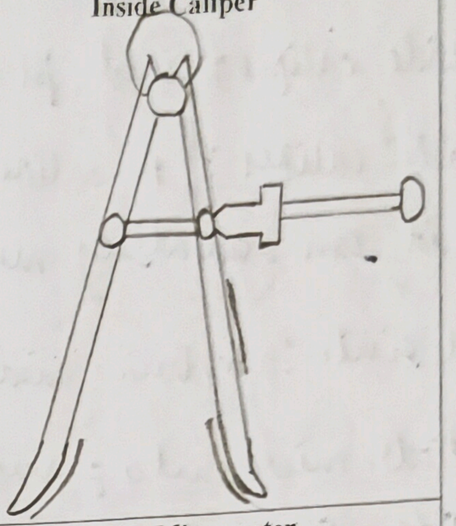
* Involute gear cutter :- used for generating tooth profile of the gear and cutter.

Sketches of Measuring Tools:

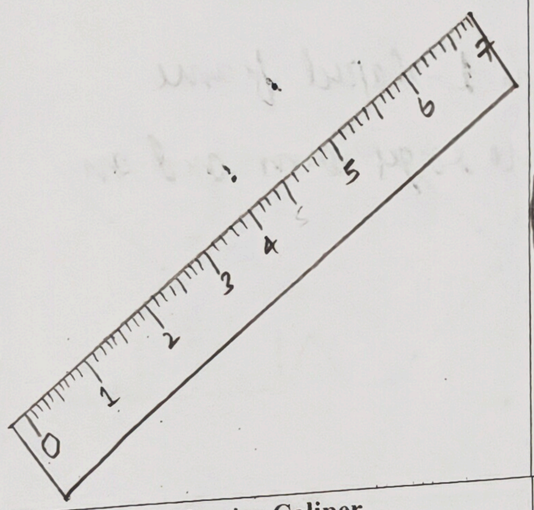
Outside Caliper



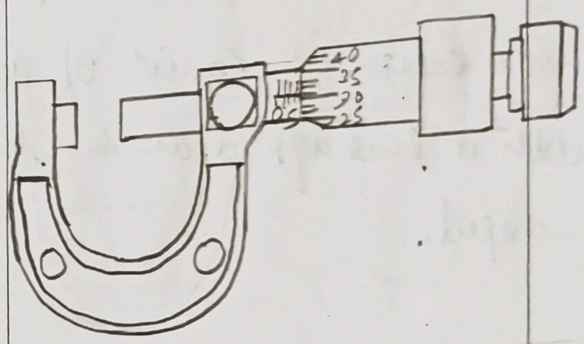
Inside Caliper



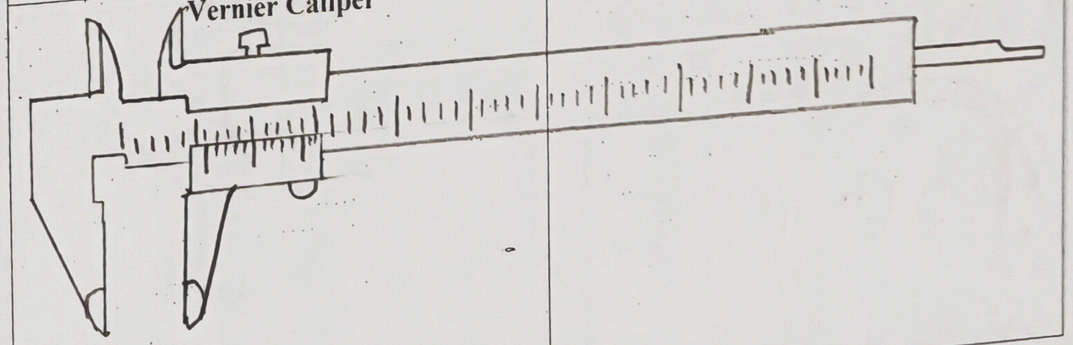
Steel Rule



Outside Micrometer



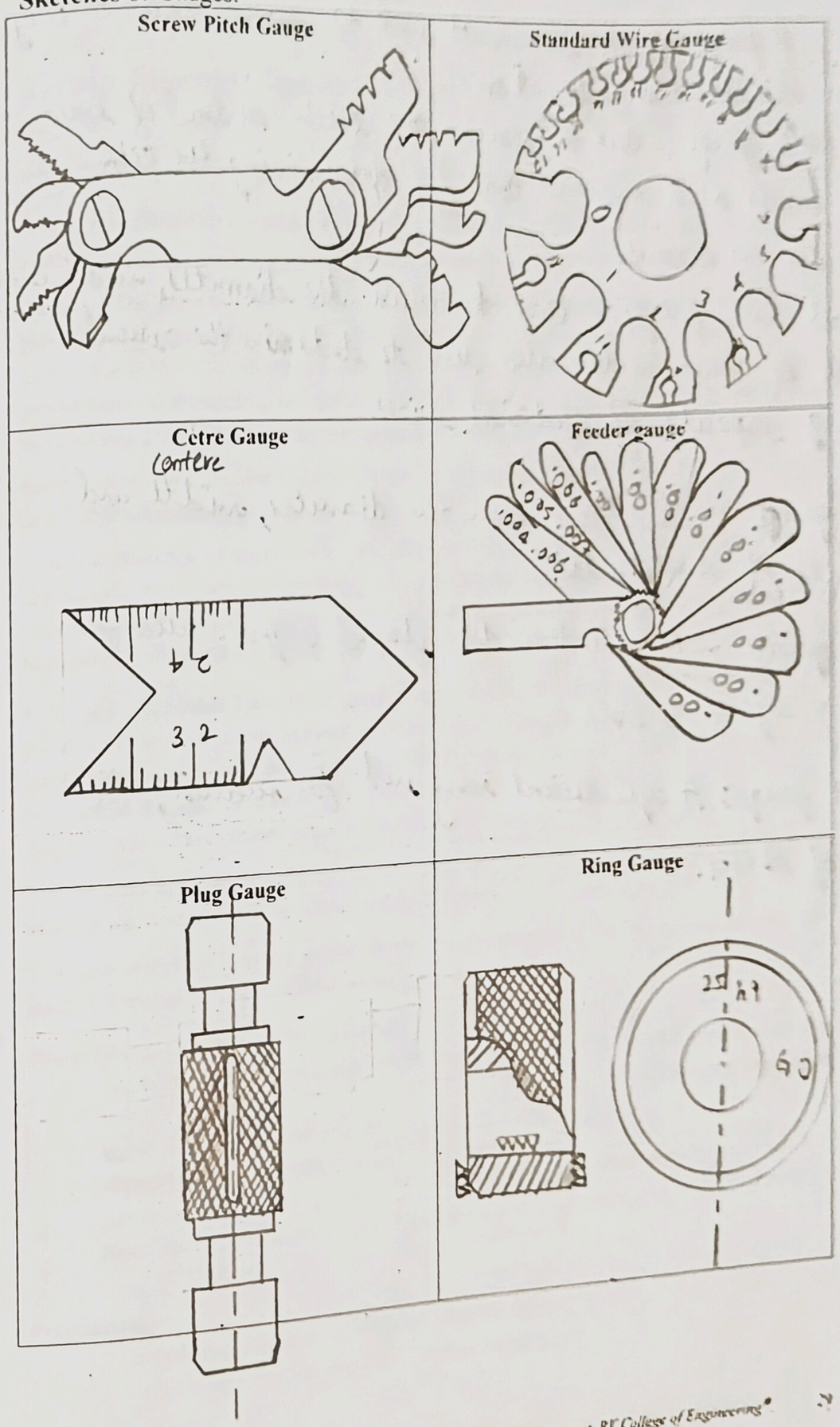
Vernier Caliper



Description of Measuring Tools:

* Outside caliper :- used for measuring external dimension.
* Inside caliper :- A caliper whose leg has outward so it can accurately measure internal dimension.
* Steel rule :- A basic measuring tool.
* Outside micrometer :- used for measuring the thickness of outside diameter of small parts.
* Vernier caliper :- Consist of an L-shaped frame with a linkage scale on the longer arm and an L-shaped.

Sketches of Gauges:



Description of Gauges:

* Gauge :- A gauge is an instrument used to measure and certain dimensional information.
* Screw pitch gauge :- Used to measure the pitch or lead of a thread, works as a reference tool for determining the pitch of threads on screws.
* Standard wire gauges :- Used to measure the diameter, width, thickness of sheet metal also used to determine the current carrying capacity of electrical wires.
* Feeler gauge :- Used to measure the diameter, width and thickness of sheet metal.
* Plug gauge :- A cylindrical shaped gauge. used for checking size of a hole.
* Ring gauge :- A cylindrical ring used for measuring the size of shaft.

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding of all tools	10		
Sketch	10		
Total Marks	20		

RV College of Engineering®, Bengaluru
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Department of Industrial Engineering & Management

OBSERVATION / DATA SHEET

Date: 28/10/2025 Expt., No: 02

Title: Determination of Grain Fineness Number (GFN) using Sieve shaker.

Observations:

Weight of the sample (Sand) = 50 grams

Sifting time = 15 minutes

Sl.No	I.S. Sieve size	Multiplier (X)	Weight Retained In Grams	% Weight Retained (Y)	Cumulative % Weight Retained	X * Y	
1	PAN	300	0.1	0.2%	0.2	~~0.6~~	60
2	53 µ	200	0.1	0.2	0.4	~~0.4~~	40
3	75 µ	140	0.1	0.2	0.6	~~0.28~~	28
4	106 µ	100	2	4	4.6	4	400
5	150 µ	70	2	4	8.6	~~2.8~~	280
6	212 µ	50	11	22	30.6	~~1~~	1100
7	300 µ	40	7	14	44.6	~~5.6~~	560
8	425 µ	30	7	14	58.6	~~4.2~~	420
9	600 µ	20	10	20	78.6	4	400
10	850 µ	10	5	10	88.6	1	100
11	1.70 mm	5	2	4	92.6	~~0.2~~	20
12	3.35 mm	3					
				92.6%	ΣX*Y =	~~24.18~~	3408

Formulae used:

1. % of sand retained in each sieve $= \frac{Weight\ of\ retained\ sand}{50} \times 100$

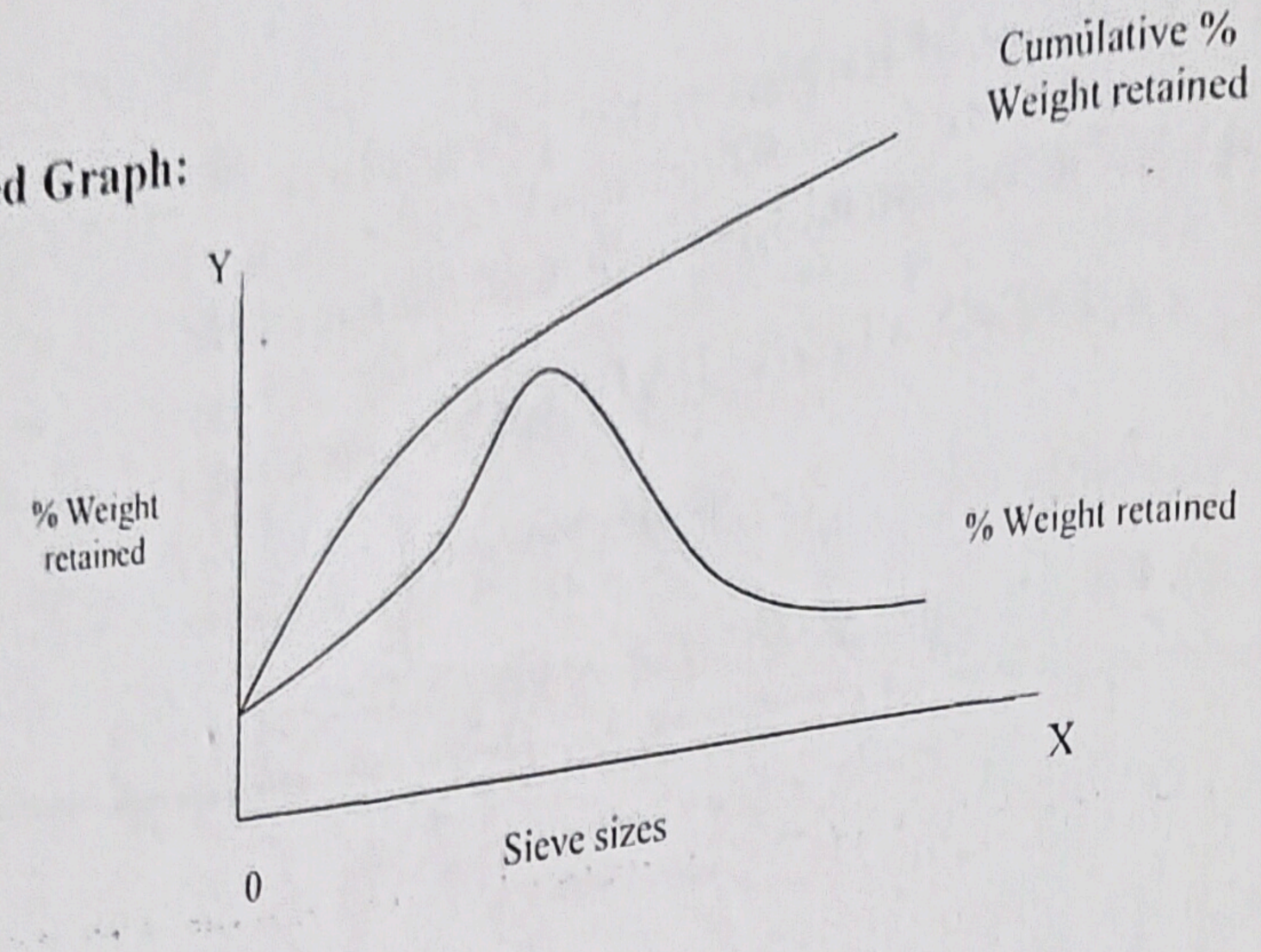
$= \frac{2}{50} \times 100$

$= 4$

2. AFS Grain fineness number $= \frac{\Sigma(\%\ of\ weight\ retained \times multipler)}{Total\ \%\ weight\ retained}$

GFN $= \frac{3408}{92.6} =$ ~~0.261~~ 36.80

Expected Graph:



Result and Discussion:

* The sieve analysis revealed a well graded distribution of gra
  size found in the sieve analysis.
* The sample is primarily composed of medium sized gra
  as evidenced by the highest % of material retained
* A small amount of fine went through the finest sieve, indi
  a low dust content and good structural stability.
* Better compaction qualities & enhanced engineering perform
  for casting use are implied by the uniform grading.

Scheme of Evaluation:

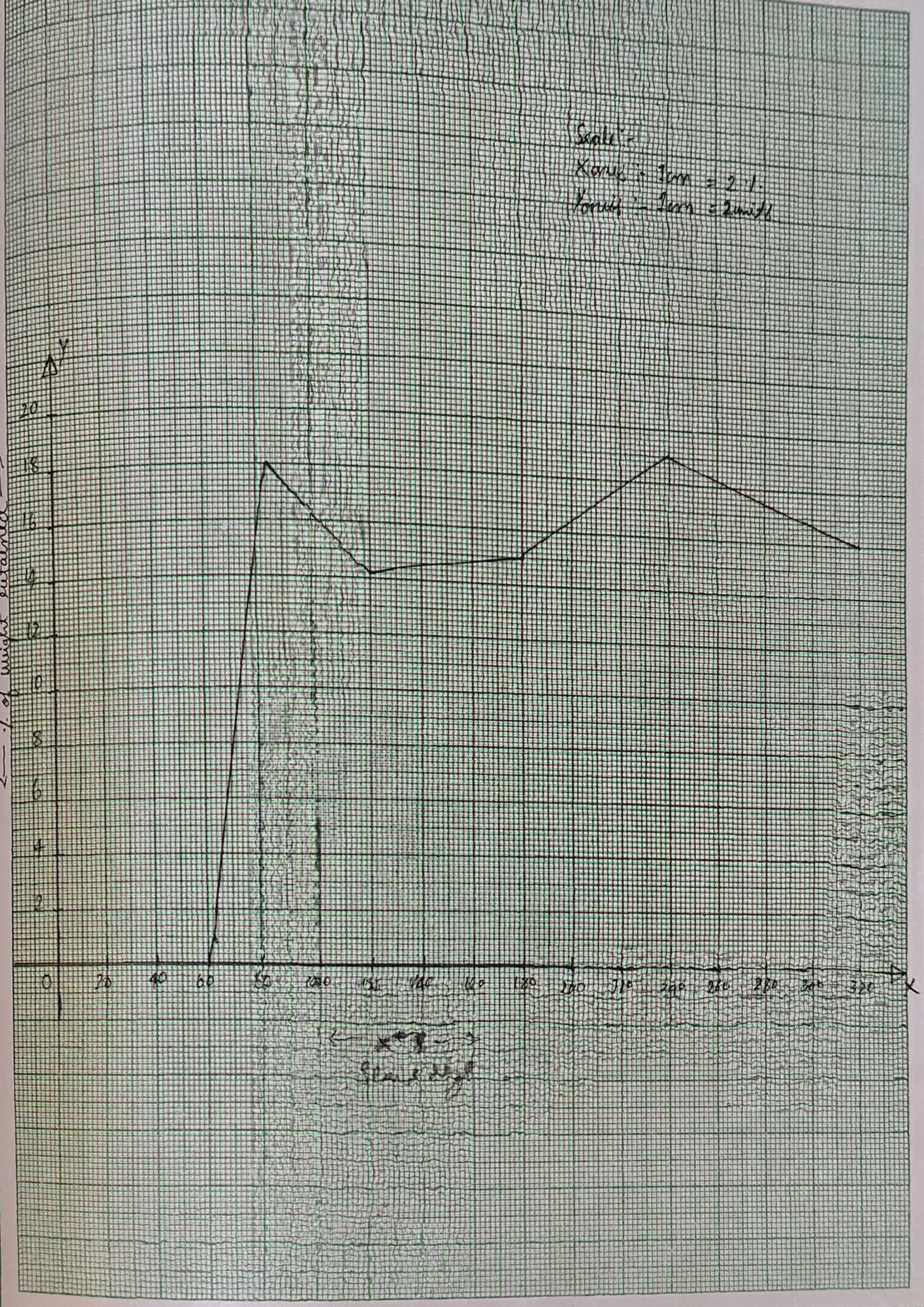
Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding	04		
Conduction of experiment, Tabulation and calculation	06		
Analysis & Inference	05		
Viva	05		
Total Marks	20		

Date : 28 10 2025

Scale :-

Xaxis : 1cm = 2 l.

Yaxis : 1cm = 2units



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Department of Industrial Engineering & Management

OBSERVATION / DATA SHEET

Date: 28/10/2025

Expt., No: 03

Title: **Evaluation of the Effect of Moisture Content on Shear Strength**

Observation:-

1. Varying the percentage of Clay:

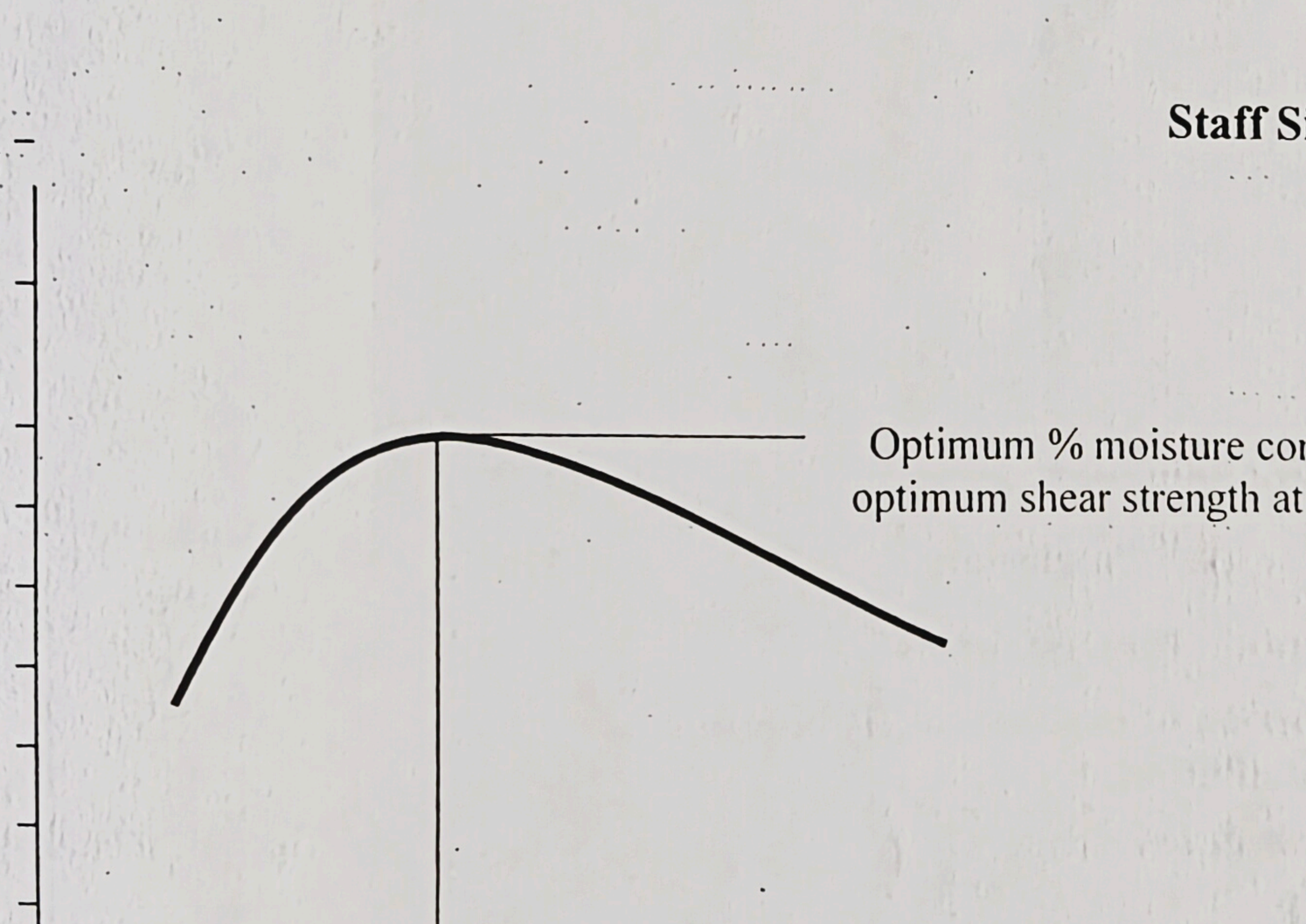
Sl. No.	Percentage of sand	Percentage of clay	Percentage of water	Shear strength gm/cm²
1.	91%(182gms)	4%(8gms)	5%(10gms)	0.64
2.	90%(180gms)	5%(10gms)	5%(10gms)	0.70
3.	89%(178gms)	6%(12gms)	5%(10gms)	0.60

2. Varying the percentage of Water:

Sl. No.	Percentage of sand	Percentage of clay	Percentage of water	Shear strength gm/cm²
1.	90%(180gms)	6%(12gms)	4%(8gms)	0.54
2.	89%(178gms)	6%(12gms)	5%(10gms)	0.76
3.	88%(176gms)	6%(12gms)	6%(12gms)	0.56

Staff Signature

Expected Graph:



Inference:

The Graphs above reveal:

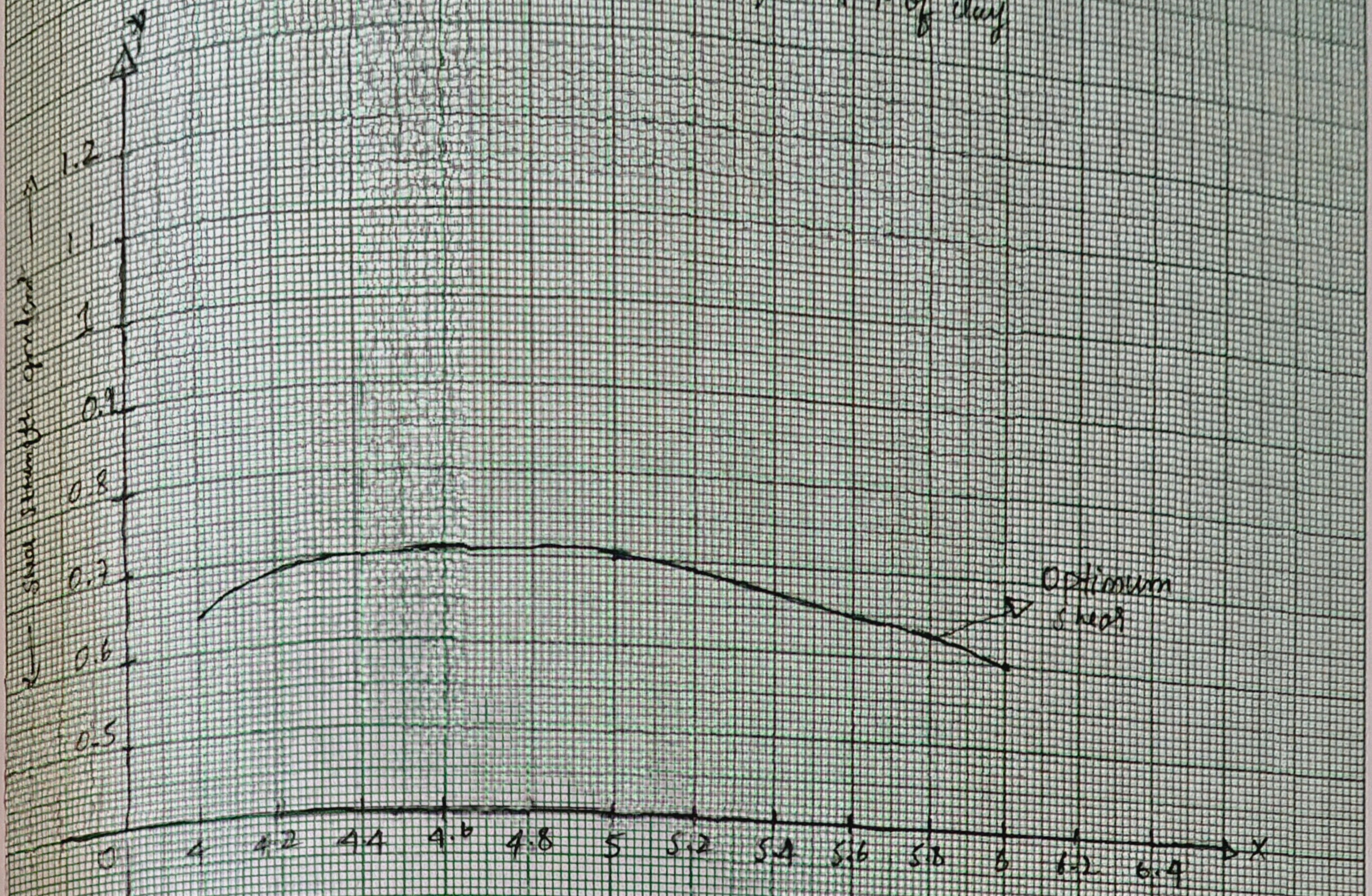
1. With the increase in the percentage of water the shear strength of the specimen initially increases up to an optimum moisture level and thereafter decreases due to the watering and lubrication effect of excess water.
2. With the increase in the percentage of clay the shear strength of the specimen increases the bonding strength blue sand grains is strengthened by a higher clay content.

Scheme of Evaluation:

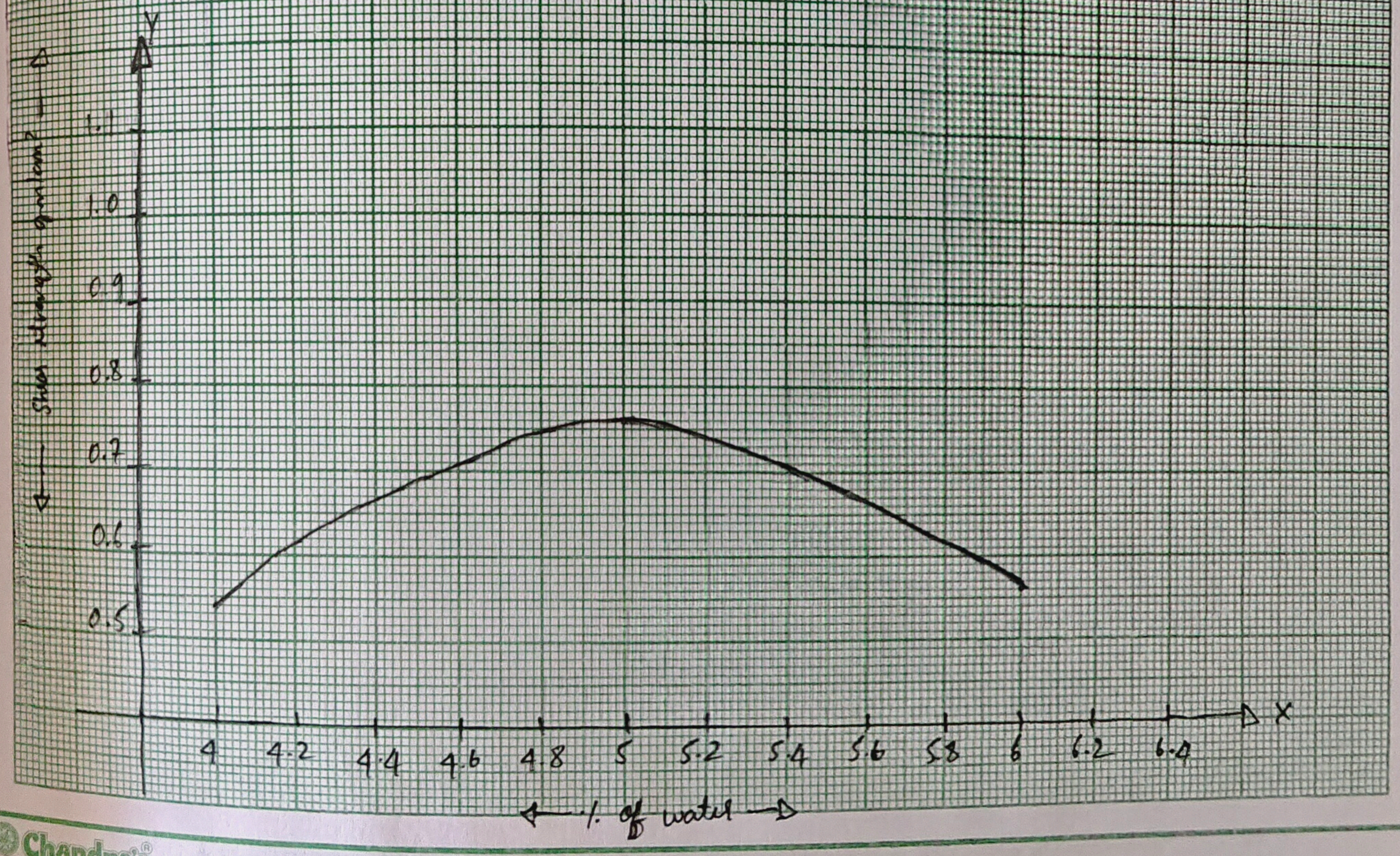
Particulars	Marks	Marks obtained	Staff Signature
Conceptual Understanding	04		
Conduction of experiment, Tabulation and calculation	06		
Analysis & Inference	05		
Viva	05		
Total Marks	20		

Date : 28 10 2025

Graph I = Shear Strength vs % of clay



Graph II = Shear Strength vs % of water

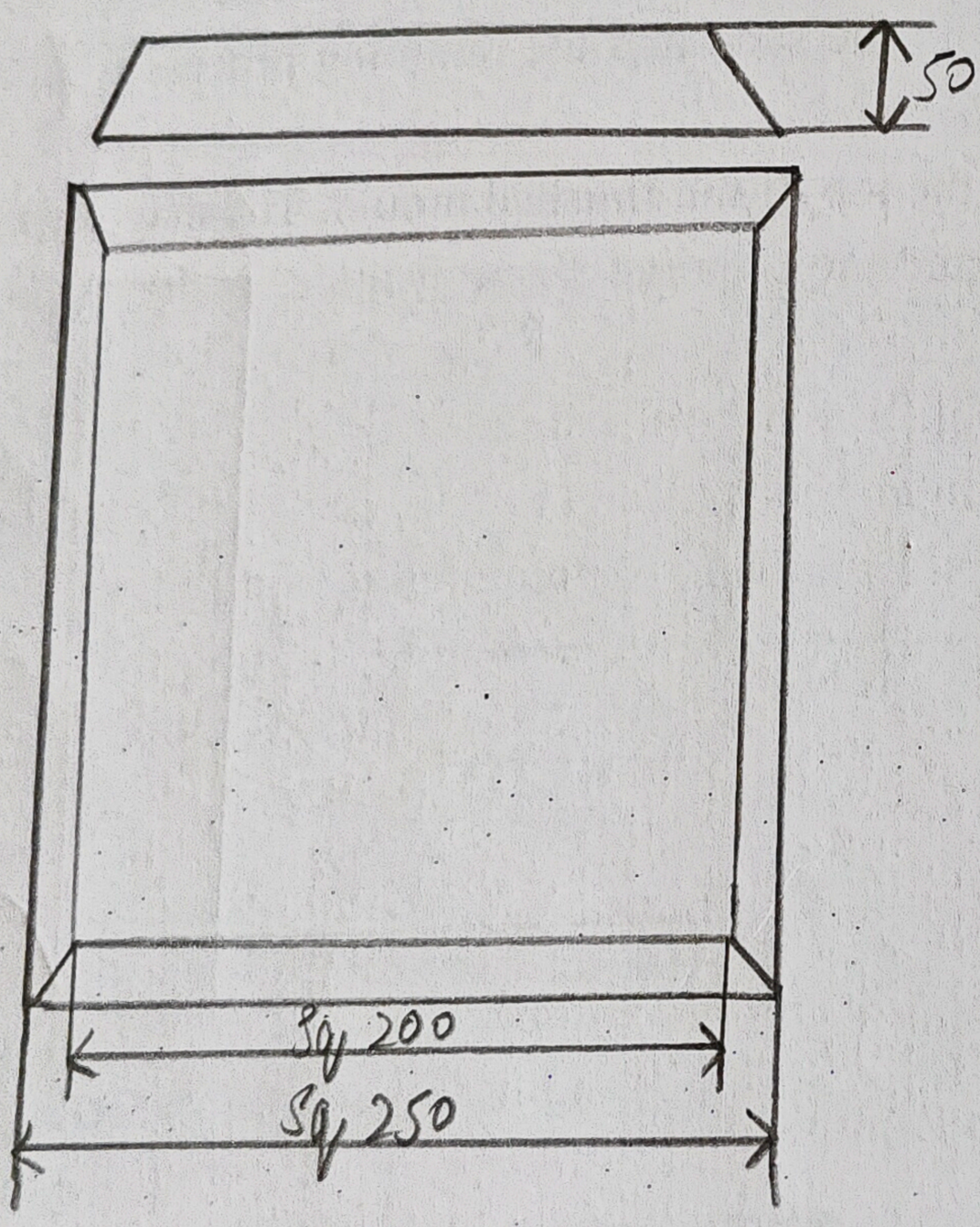


Chandra's®

Sequence of Operations:

* Mix the casting sand
* Place the cope and drag & fill the sand
* Edge the sand using rig lammers
* Ram sand lammig and lammig equally.
* Remove the top flask
* Mark the middle line
* Mark the core dimensions.
* Cut the core vents in taper
* Ill trowell and give finishing.

Model with Actual Dimension in mm



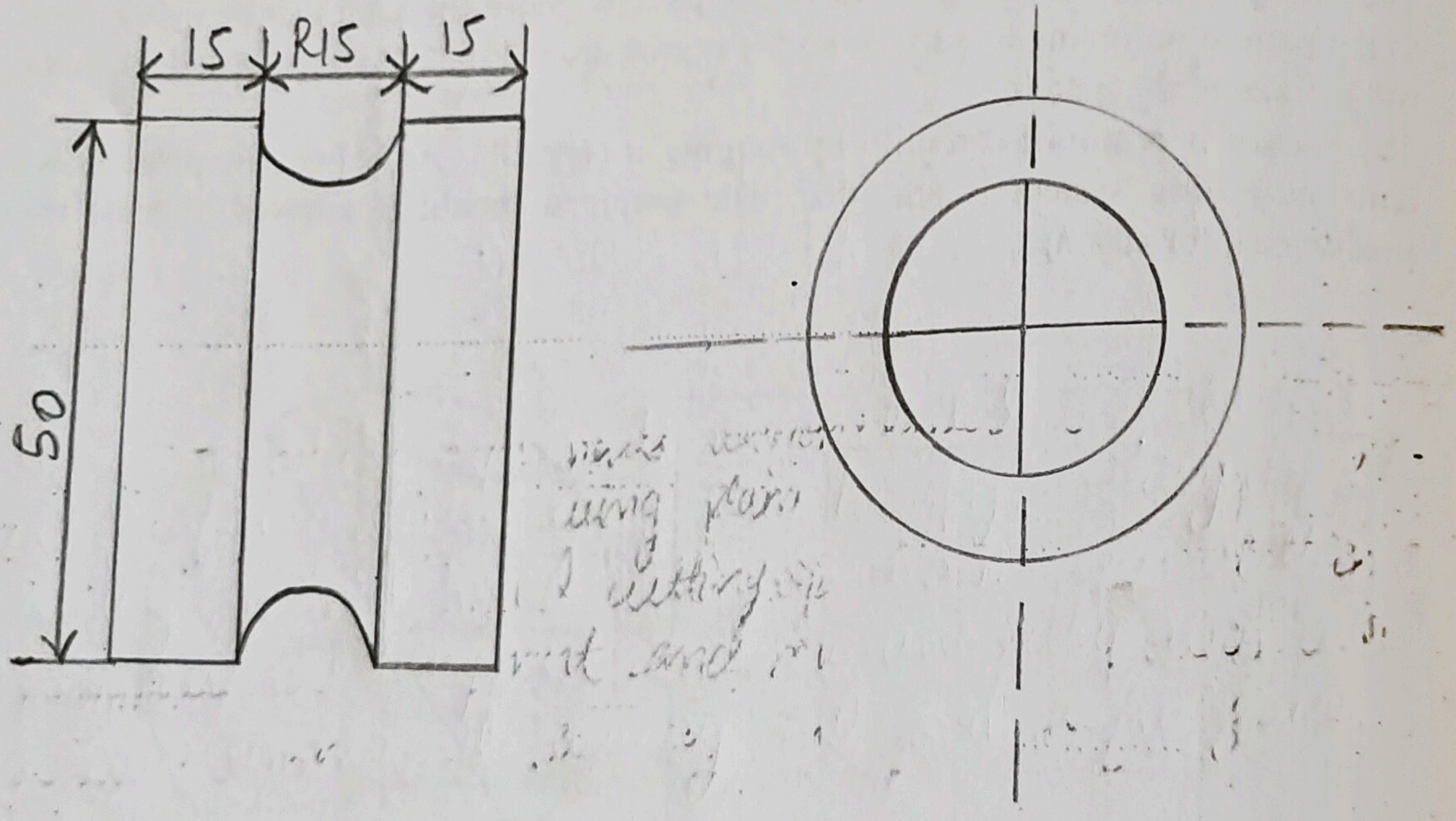
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**08**		
Finish	**08**		
Viva	**04**		
Total Marks	**20**		

Sequence of Operations:

* Place the drag upside down, place the split pattern
* Fill the sand and ram properly ram the edges
* Upside down the drag and place cope above it, keep the another half of split pattern on the another.
* Apply parting sand
* Fill the sand by sieving remove & fill.
* Ram equally.
* Split the cope & drag and remove runner, split pattern.
* Use trowel and give finishing.

Model with Actual Dimension:



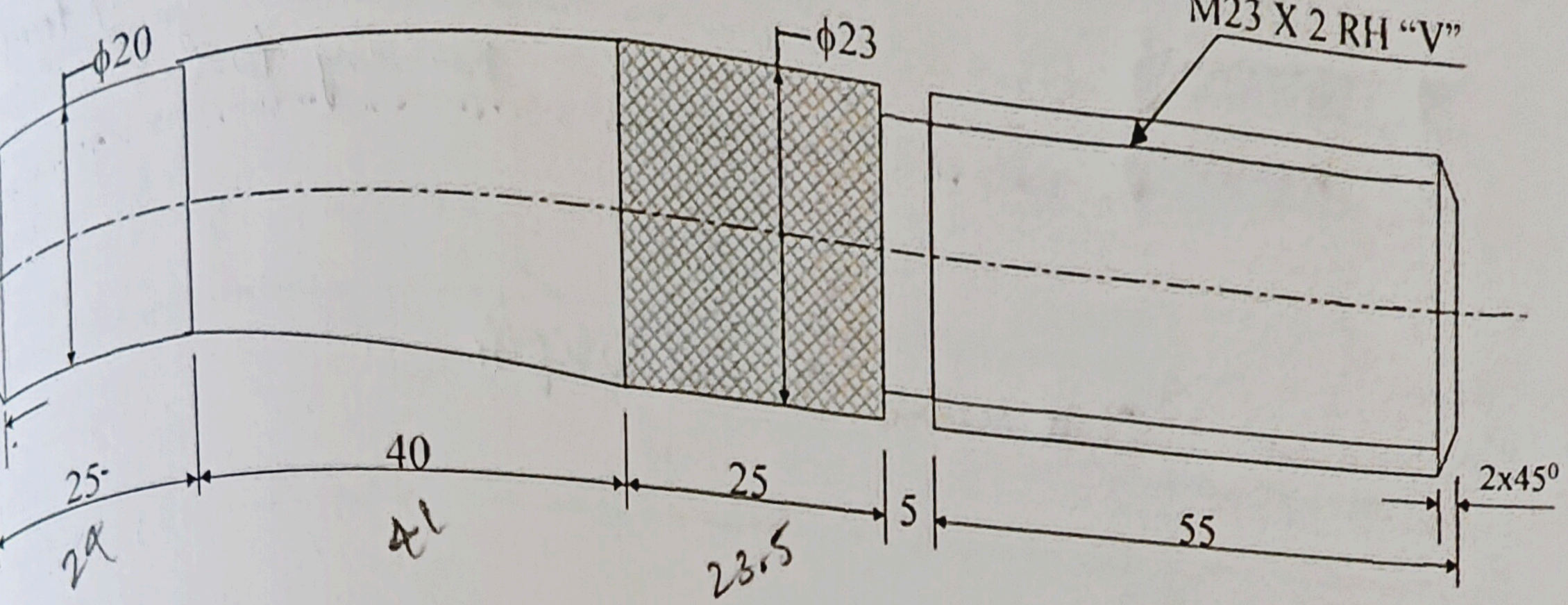
Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	08		
Finish	08		
Viva	04		
Total Marks	20		

Experiment

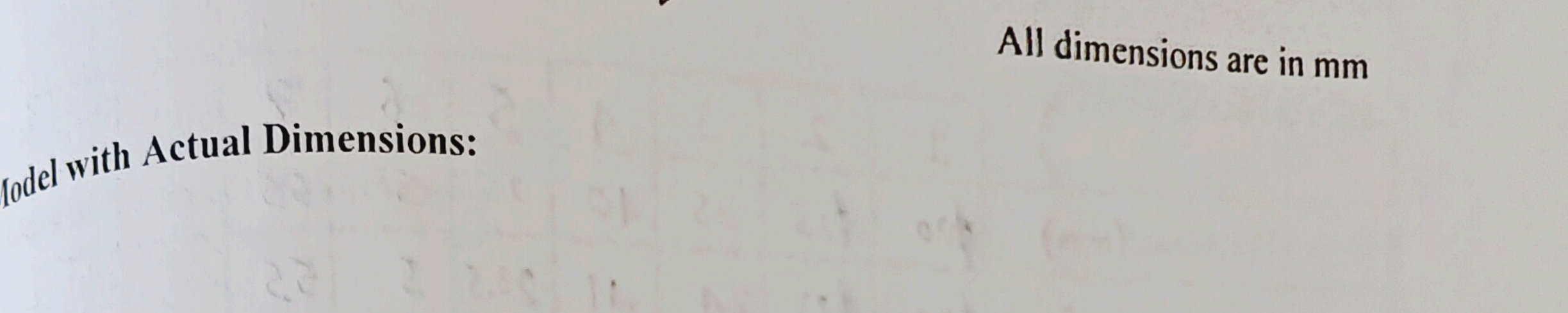
06

ain Turning, Grooving, Taper turning, Knurling,
xternal Thread Cutting



All dimensions are in mm

Model with Actual Dimensions:



Taper angle calculation :- Given: D = 23 mm
d = 20 mm
L = 40 mm

$$TAF = \tan\theta = \frac{D-d}{2L} = \frac{23-20}{2\times40}$$

$$= 0.0375$$

$$\theta = \tan^{-1}(0.0375)$$

$$\therefore \theta = 2.15^{\circ}$$

Operation Plan:

Sl No	Sequence of Operations	Tools Used
1	Plain turning	V - Tool
2	Step turning	V - Tool
3	Tapes turning	V - Tool / Grooving tool
4	Grooving	Knurling tool
5	Knurling	
Measuring Instruments Used	Steel rules, Vernier Calipers.	

	1	2	3	4	5	6	7		
Given Dimension (mm)	ϕ20	ϕ23	25	40	25	5	55		
Measured Dimension (mm)	ϕ20	ϕ23	24	41	23.5	5	55		
Deviation (mm)	0	0	1	-1	1.5	0	0		

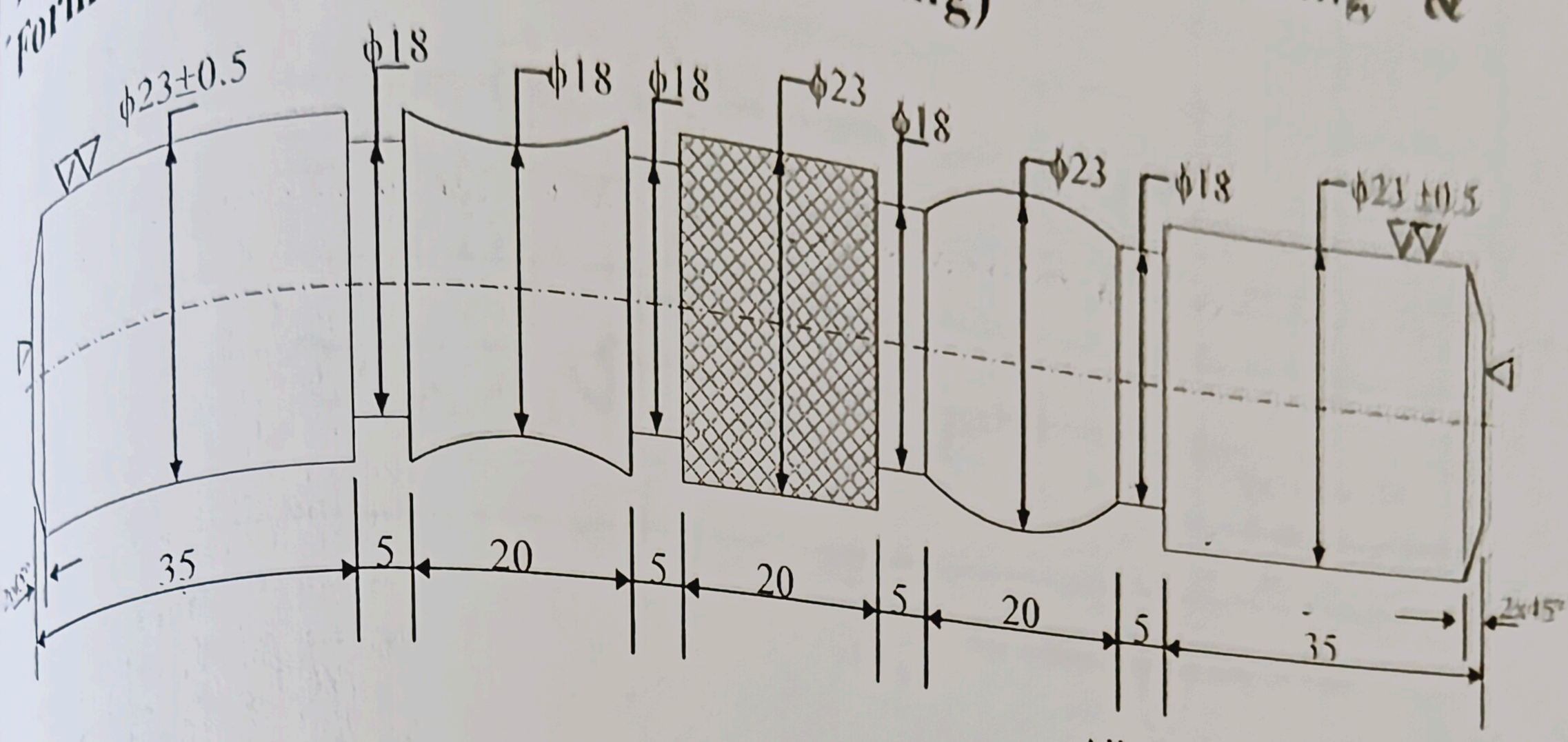
Inference: The various experiments demonstrated accurate material removal and surface generation using plain turning, grooving, taper turning, step, knurling & external thread cutting operations. Proper selection of cutting parameters, tool alignment and machine setup was essential to achieve dimensional accuracy and good surface finish.

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	**05**		
Surface finish	**05**		
Report writing	**05**		
Viva	**05**		
Total Marks	**20**		

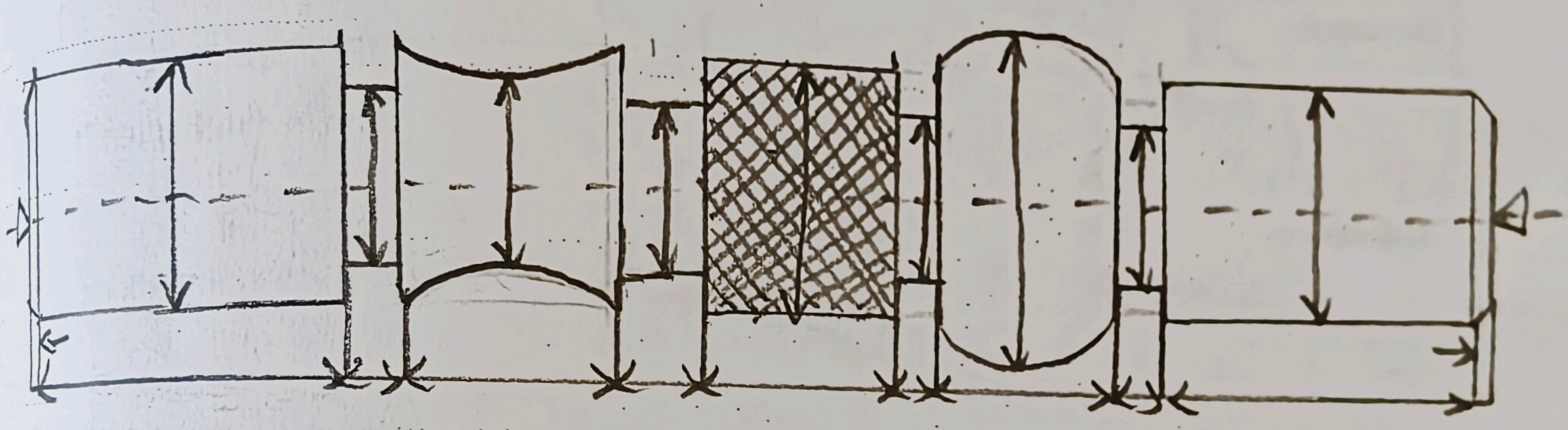
Experiment
07

plain Turning, Grooving, Knurling, Chamfering &
Forming (Concave & Convex Turning)



All dimensions are in mm

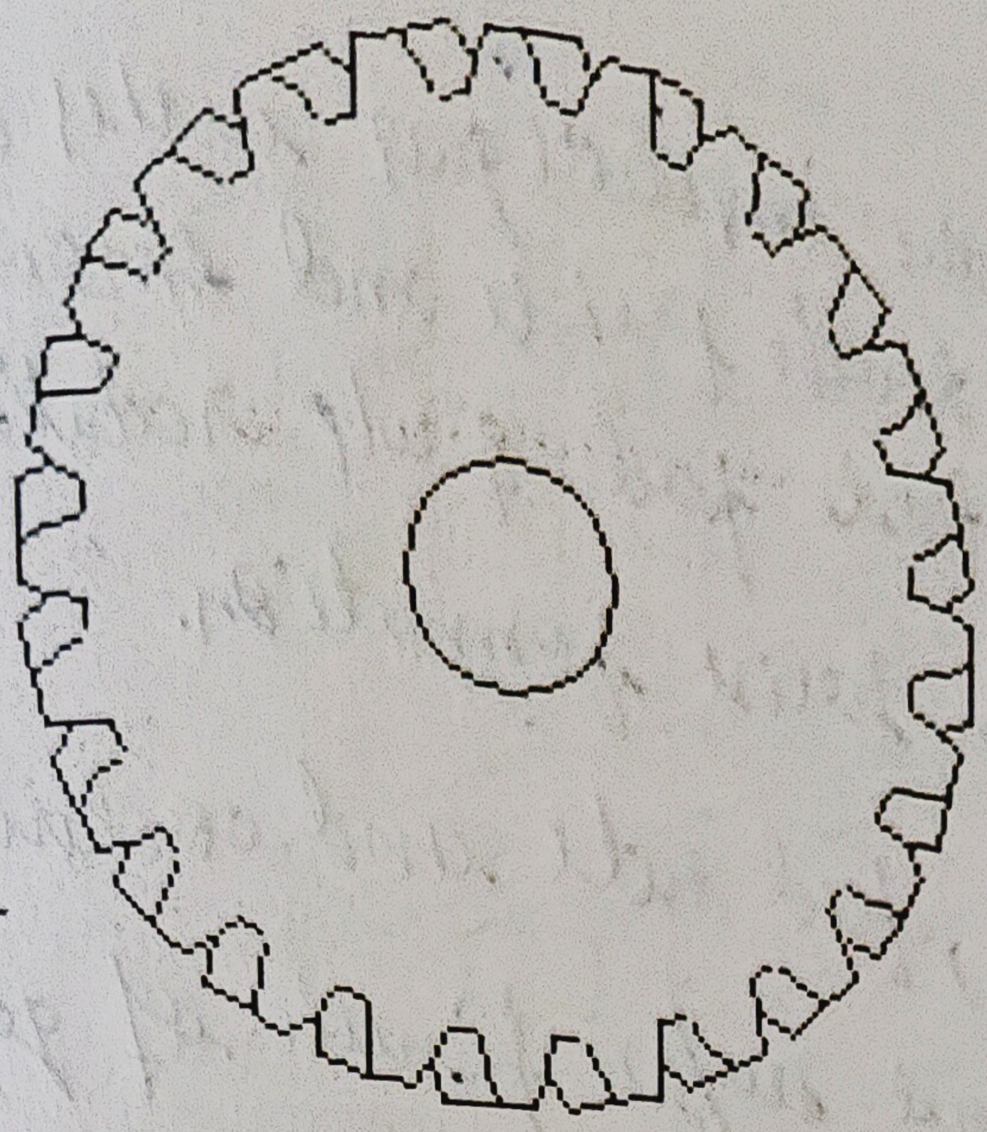
Model with Actual Dimensions:

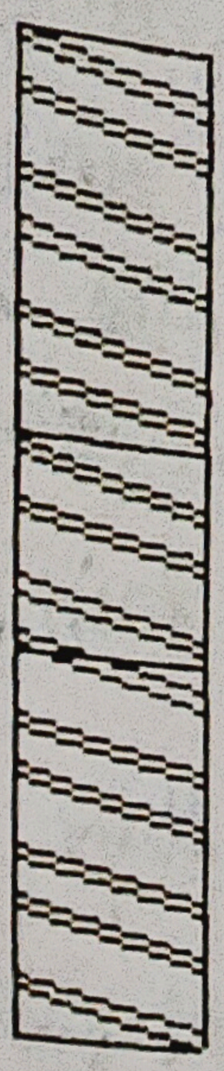


Experiment

09

Helical Gear Cutting using Milling Machine





Calculations:

1. Outer diameter $= m[Z+2]$ $= 2.5(2+Z) =$ 78mm

   Z = no of teeth $Z = 29.2 \simeq 29$

2. Pitch Circle diameter $= mZ$

   m = 2.5

   $= 2.5 \times 29 \simeq 72.5$

3. Circular Pitch $= \pi m$

   m = 2.5

   $= \pi \times 2.5 =$ 7.85

4. Working Depth $= 2.25\ m$

   $= 2.25 \times 2.5 =$ 5.625 mm

5. Indexing $= \dfrac{40}{Z}$ $= 1.3793 = 11\dfrac{1}{29}$

6. Tan $\theta = \dfrac{\pi d}{l}$ 20.55 where L = 450

   $\theta =$ 28.57

7. Change gear ratio $= \dfrac{\text{Lead of the machine (l)}}{\text{Lead to be cut (L)}} = \dfrac{200}{450} = \dfrac{4}{9} \times \dfrac{8}{8} =$ 0.44

24, 24, 26, 28, 32, 36, 40, 46, 48, 72, 76, 86, 100

Inference:

* Helical gear cutting was succesfully performed by properly indexing the gear blank and setting the correct helix angle on the milling m/c.
* Accurate selection of the involute gear cutter and proper depth of cut ensured correct tooth profile and smooth meshing.
* Synchronization b/w table feed & w/p rotation was crucial to maintain uniform helix formation.
* Proper cutting speed, feed rate and coolant usage reduced tool wear and improve surface finish of gear teeth.
* The experiment helped to understand of gear geometry, indexing mechanisms and precision control in milling operations.

Scheme of Evaluation:

Particulars	Marks	Marks obtained	Staff Signature
Dimensional Accuracy	05		
Surface finish	05		
Report writing	05		
Viva	05		
Total Marks	20		